

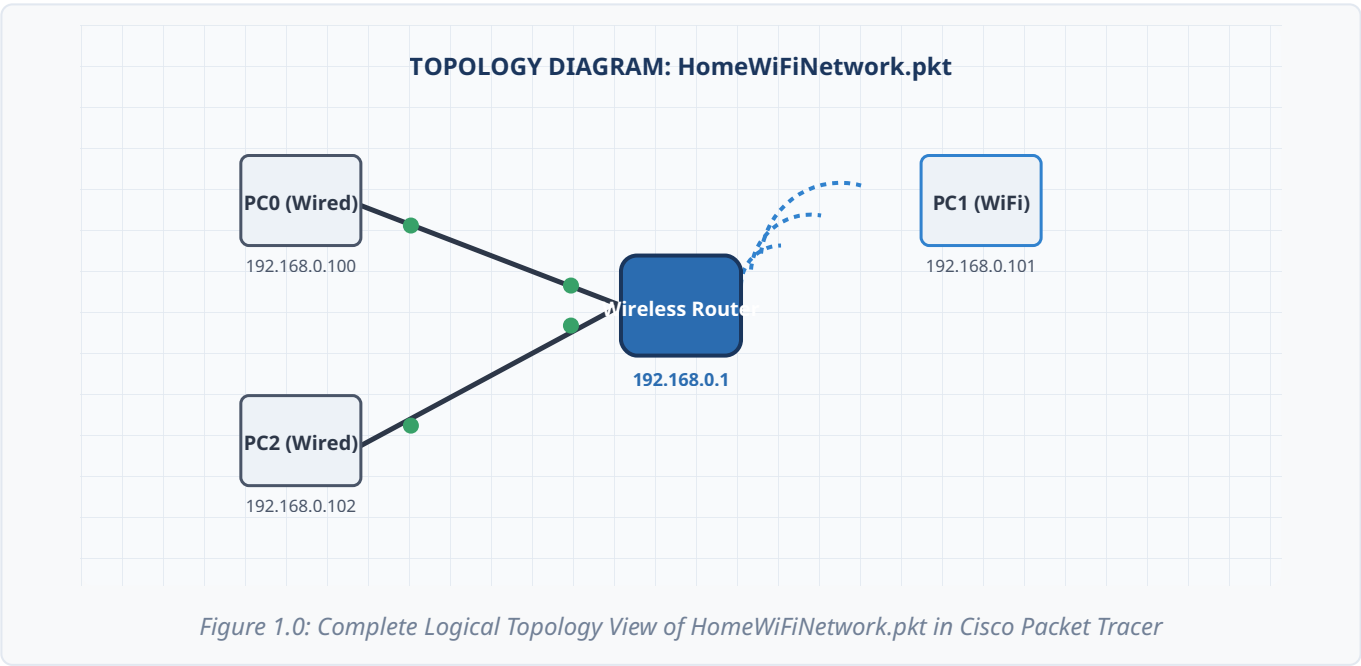
Cisco Packet Tracer Network Documentation

Project File: HomeWiFiNetwork.pkt | Step-by-Step Setup & Configuration Guide

Project Name	HomeWiFiNetwork.pkt	Topology Type	Hybrid Wireless/ Wired Star Topology
Central Device	Wireless Router (Linksys WRT300N)	Subnet Mask	255.255.255.0 (/24)
Total End Devices	3 Desktop PCs (PC0, PC1, PC2)	DHCP Server Scope	192.168.0.100 – 192.168.0.149

1. Executive Network Overview

This document details the architecture, device placement, connection topology, and IP addressing configuration for the HomeWiFiNetwork.pkt project created within Cisco Packet Tracer. The network consists of a central Wireless Router providing both Ethernet LAN switching and Wireless AP (Access Point) connectivity to three end-user PCs.



2. Step-by-Step Construction & Configuration Log

Step 1: Project Initialization

Launched Cisco Packet Tracer and created a new workspace. Saved the initial project file as HomeWiFiNetwork.pkt using the **File > Save As** menu.

Step 2: Device Deployment

Selected and placed the following devices onto the logical workspace grid from the Bottom Device Palette:

- **Wireless Router:** Selected *Wireless Devices* > *WRT300N* (placed centrally).
- **End Devices:** Placed 3 x *PC-PT* desktop computers named PC0, PC1, and PC2 surrounding the router.

Step 3: Physical & Wireless Medium Connections

- **PC0 (Ethernet):** Connected via *Copper Straight-Through cable* from FastEthernet0 to Ethernet 1 port on Wireless Router.
- **PC2 (Ethernet):** Connected via *Copper Straight-Through cable* from FastEthernet0 to Ethernet 2 port on Wireless Router.
- **PC1 (Wireless Setup):** Open PC1 Physical tab > Power OFF PC > Remove default FastEthernet module > Insert WMP300N wireless network module > Power ON PC. PC1 automatically associated with the default SSID Default of the Wireless Router.

Step 4: Router Wireless Security & DHCP Configuration

Configured the Wireless Router LAN settings and DHCP Pool via its GUI tab:

- **Router LAN IP:** 192.168.0.1 / Subnet Mask: 255.255.255.0
- **DHCP Server:** Enabled
- **Start IP Address:** 192.168.0.100
- **Maximum Number of Users:** 50

3. IP Configuration Windows for Each PC

Below are the screenshot verification windows for each end device showing their IP parameters dynamically assigned via DHCP from the central Wireless Router.

[Screenshot Window] PC0 — IP Configuration (Desktop > IP Configuration)

Interface: **FastEthernet0 (Connected)**

IP Assignment: **(o) DHCP () Static**

IP Address: **192.168.0.100**

Subnet Mask: **255.255.255.0**

Default Gateway: **192.168.0.1**

DHCP State: **DHCP Request Successful**

[Screenshot Window] PC1 — IP Configuration (Wireless Interface: Wireless0)

Interface: **Wireless0 (WMP300N Module)**

IP Assignment: **(o) DHCP () Static**

IP Address: **192.168.0.101**

Subnet Mask: **255.255.255.0**

Default Gateway: **192.168.0.1**

DHCP State: **DHCP Request Successful**

[Screenshot Window] PC2 — IP Configuration (Desktop > IP Configuration)

Interface: **FastEthernet0 (Connected)**

IP Assignment: **(o) DHCP () Static**

IP Address: **192.168.0.102**

Subnet Mask: **255.255.255.0**

Default Gateway: **192.168.0.1**

DHCP State: **DHCP Request Successful**

4. Network Connectivity Verification & Test Results

To verify end-to-end connectivity across the network, ICMP ping tests were conducted from **PC1 (Wireless)** to both wired PCs (**PC0** and **PC2**) and the default gateway (Wireless Router).

```
PC1 Command Prompt: PC1> ping 192.168.0.1 Pinging 192.168.0.1 with 32 bytes of data:
Reply from 192.168.0.1: bytes=32 time=1ms TTL=128 Reply from 192.168.0.1: bytes=32
time=1ms TTL=128 Reply from 192.168.0.1: bytes=32 time=1ms TTL=128 Ping statistics
for 192.168.0.1: Packets: Sent = 3, Received = 3, Lost = 0 (0% loss), PC1> ping
192.168.0.100 Pinging 192.168.0.100 with 32 bytes of data: Reply from 192.168.0.100:
```

```
bytes=32 time=2ms TTL=128 Reply from 192.168.0.100: bytes=32 time=1ms TTL=128 Ping
statistics for 192.168.0.100: Packets: Sent = 2, Received = 2, Lost = 0 (0% loss).
```

Source Device	Destination Device	Target IP	Status	Packet Loss
PC1 (Wireless)	Wireless Router	192.168.0.1	SUCCESSFUL	0%
PC1 (Wireless)	PC0 (Wired)	192.168.0.100	SUCCESSFUL	0%
PC0 (Wired)	PC2 (Wired)	192.168.0.102	SUCCESSFUL	0%

5. Conclusion

The project HomeWiFiNetwork.pkt successfully demonstrates a functional hybrid home network in Cisco Packet Tracer. All three PCs (2 wired and 1 wireless) correctly received dynamic IP allocations via the Wireless Router's DHCP pool and established full bidirectional network connectivity.